Consider the following problem

$$\min_x \sum_{i\in I} f_i(x_i) \quad s.t. \quad \begin{cases} g_i(x_i) = 0\,, \quad i \in \{1,\dots,N\} \\ \sum_{i\in I} A_i x_i = b \end{cases}$$

where $x_i \in \mathbb{R}^{n_x}$, $f_i : \mathbb{R}^{n_x} \to \mathbb{R}$, $g_i : \mathbb{R}^{n_x} \to \mathbb{R}^{n_g}$, and $A_i \in \mathbb{R}^{n_A \times n_x}$

Our aim is to exploit the structure of problem to distribute computations on local level as much as possible, as opposed to decomposing the problem,

If we associate multipliers $\lambda_i \in \mathbb{R}^{n_g}$ with g_i and $\mu \in \mathbb{R}^{n_A}$, we will have the following Lagrangian and its derivatives:

$$L(x,\lambda,\mu) = \sum_{i\in I} f_i(x_i) + \lambda_i^T g_i(x_i) + \mu^T (\sum_{i\in I} A_i x_i - b)$$

$$\nabla_x L(x,\lambda,\mu) = \begin{bmatrix} [\nabla_x L(x,\lambda,\mu)]_1 \\ \vdots \\ [\nabla_x L(x,\lambda,\mu)]_N \end{bmatrix}$$

and

$$[\nabla_x L(x,\lambda,\mu)]_i = \nabla_{x_i} f_i(x_i) + \nabla_{x_i} g_i(x_i)^T \lambda_i + A_i^T \mu$$

where ∇g is the Jacobian of the corresponding function.

Therefore to find the minimizer we solve the following problem;

$$\nabla_x L(x,\lambda,\mu) = 0 \quad s.t. \quad \begin{cases} g_i(x_i) = 0\,, \quad i \in \{1,\dots,N\} \\ \sum_{i\in I} A_i x_i = b \end{cases}$$

Presuming regularity, and denoting vector of (x_i,λ_i) as z_i, solving this problem using the Newton method will result in the following system of equations.

$$\begin{bmatrix} KKT \end{bmatrix} \begin{bmatrix} \Delta z \\ \Delta \mu \end{bmatrix} = - \begin{bmatrix} [\nabla_x L(x,\lambda,\mu)]_i \\ g_i(x_i) \\ \vdots \\ \sum_{i\in I} A_i x_i - b \end{bmatrix} = - \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} \tag{1}$$

Where Δz is a vector of size $N \times (n_x + n_g)$,and $\Delta \mu$ of size n_A with the following KKT matrix;

$$KKT = \begin{bmatrix} \begin{bmatrix} \ddots & & \\ & \begin{bmatrix} \nabla^2_{x_i} f_i(x_i) + < \lambda_i, \nabla^2_{x_I} g_i(x_i) > & \nabla_{x_i} g_i(x_i)^T \\ \nabla_{x_i} g_i(x_i) & 0 \end{bmatrix} & \\ & & \ddots \end{bmatrix} & \begin{bmatrix} \vdots \\ A_i^T \\ 0 \\ \vdots \end{bmatrix} \\ \begin{bmatrix} \cdots & A_i & 0 & \cdots \end{bmatrix} & \begin{bmatrix} 0 \end{bmatrix} \end{bmatrix}$$

The matrix containing Hessian of the Lagrangian and Jacobians of fucntions g_i's consists of N blocks of size $(n_x + n_g) \times (n_x + n_g)$ (with the zero matrix being the size $n_g \times n_g$), and the matrix $\bar{A}_i$ consisting of $[A_i, 0]$ has N matrices of size $n_A \times (n_x + n_g)$.

Let us denote the block matrix containing the Hessian of the Lagrangian and the Jacobians as matrix $\mathbf{D}$ and each sub matrix by D_i, the block containing matrices $\bar{A}_i$ as $\bar{\mathbf{A}}$, the vector containing i-th component of $\nabla_x L(x, \lambda, \mu)$ and $g_i(x_i)$ as r_1, and $\sum_{i \in I} A_i x_i - b$ as r_2 in the right hand side of 1. Then we can solve for ΔZ:

$$\begin{aligned} D\Delta z + \bar{\mathbf{A}}^T \Delta\mu &= -r_1 \\ \Delta z &= -D^{-1} r_1 - D^{-1} \bar{\mathbf{A}}^T \Delta\mu \end{aligned}$$

Where due to the structure of the KKT matrix with $\Delta\mu$ provided can be solved for each z_i separately.

$$\Delta z_i = -D_i^{-1} r_{1_i} - D_i^{-1} \bar{\mathbf{A}}_i^T \Delta\mu$$

And

$$\bar{\mathbf{A}} \Delta z = -r_2$$

which can be written as

$$\sum_{i \in I} \bar{\mathbf{A}}_i \Delta z_i = -r_2$$

$$\sum_{i \in I} -\bar{\mathbf{A}}_i D_i^{-1} r_{1_i} - \bar{\mathbf{A}}_i D_i^{-1} \bar{\mathbf{A}}_i^T \Delta\mu = -r_2$$

$$\sum_{i \in I} -\bar{\mathbf{A}}_i D_i^{-1} \bar{\mathbf{A}}_i^T \Delta\mu = -r_2 + \sum_{i \in I} \bar{\mathbf{A}}_i D_i^{-1} r_{1_i} \tag{2}$$

where the terms with index i will be calculated locally and communicated for finding Lagrange multiplier of the coupled constraint. The updated $\Delta\mu$ will be communicated to local agents to solve for Δz.

To ensure that the calculated step yields an improvement (reduce the cost function and violation of the constraints) we choose step-size by using backtracking linesearch on a suitably defined merit function. Here we use the following l_1 merit function.

$$\phi(x) = \sum_{i \in I} f_i(x_i) + \sigma(\|g(x)\|_1 + \left\| \sum_{i \in I} A_i x_i - b \right\|_1)$$

where the value of σ has to be kept greater then the values of the Lagrange multipliers, $\sigma \geq \|(\lambda, \mu\|_\infty$.
We use Armijo coindition that accepts a step-size if:

$$\phi(x + \alpha\Delta x) \leq \phi(x) + \alpha\gamma D\phi(x)$$

where $D\phi(x)$ is the directional derivative of $\phi(x)$ in the direction of ΔX;

$$D\phi(X, s) = \sum_{i \in I} f_i(x_i) - \sigma(\|g(x)\|_1 + \left\| \sum_{i \in I} A_i x_i - b \right\|_1)$$

We will separate the evaluation of ϕ and $D\phi$ between local variables. We define $\Phi(\alpha) := \phi(x + \alpha\Delta x)$.

The following algorithms demonstrate the procedure.

Algorithm 1 Basic distributed algorithm for linear quadratic problem

Input: Problem parameters F_i, g_i, A_i, and b_i
Output: x_i, μ, and λ
1: Initialize x, μ, and λ
2: Locally evaluate ∇L_i, g_i, $A_i x_i$, Hessian and form matrix D_i
3: Factorize D_i and form $\bar{\mathbf{A}}_i D_i^{-1} \bar{\mathbf{A}}_i^T$, and $\bar{\mathbf{A}}_i D_i^{-1} r_{1_i}$
4: Communicate $A_i x_i$, $\bar{\mathbf{A}}_i D_i^{-1} \bar{\mathbf{A}}_i^T$, and $\bar{\mathbf{A}}_i D_i^{-1} r_{1_i}$ to central node
5: **loop**
6:     **If** KKT violation smaller than tolerance
7:         **Return** solution found
8:     **end if**
9:     Compute $\Delta\mu$ using 2 and pass it to local agents
10:     Compute Δx, and $\Delta\lambda$ locally
11:     Compute and assemble $\Phi(0)$ and $D\Phi(0)$ using local information f_i, g_i, $A_i x_i$, and b
12:     **loop**
13:         Using α, assemble $\Phi(\alpha)$
14:         **If** $\Phi(\alpha) < \Phi(0) + \alpha\gamma D\Phi(0)$ **then**
15:             **return** α
16:         **else**
17:             $\alpha \leftarrow \beta\alpha$
18:         **end if**
19:     **end loop**
20:     Update x, λ, and μ using α
21:     Recompute residuals and functions with new values, go through steps 2, 3, 4
22: **end loop**